DTS 101 From Roots to Trees, aka Devicetree for Beginners

Open Source Summit Europe 2025

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Introduction

- Krzysztof Kozlowski
- I work for Linaro
- I am the co-maintainer (with Rob and Conor) of Devicetree bindings in the Linux kernel
  - I also maintain other Linux kernel pieces
    - Memory controller drivers, NFC, 1-Wire subsystem
    - Samsung Exynos SoC ARM/ARM64 architecture
- I wrote a lot of Devicetree bindings and DTS, and I read even more

Caveats

- Through the presentation some parts will be **marked with bold and color font**. These represent common issues observed in DTS and things to watch out.
- For simplification I will use only “Linux kernel” term, but this should be understood as any upstream OS or software using the kernel’s Devicetree
  - E.g. FreeBSD, U-Boot
- Devicetree bindings are not limited to the Linux kernel and other upstream users should be considered as well
  - However the talk will not cover non-Linux kernel DTS style / rules
- The talk is considered introductory, thus obviously incomplete
  - For more rules please refer to writing-bindings.rst, writing-schema.rst and DT submitting-patches.rst in the Linux kernel sources

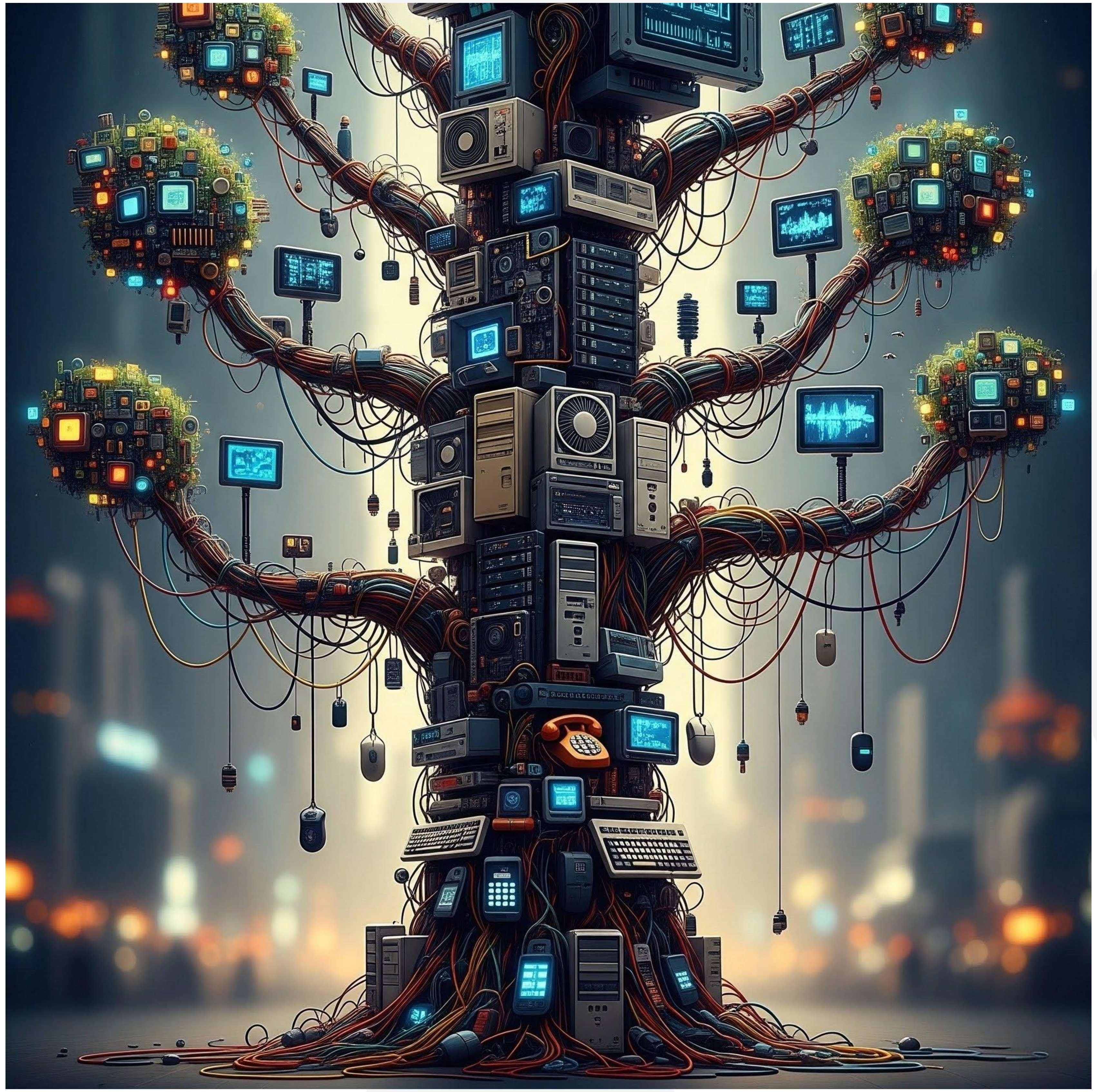
Agenda

- What is the Devicetree?
- How does a DTS look like?
- What does compatibility between devices mean and how to express it?
- How does a DT binding look like?
- What can be in DTS and what cannot
- Validate your DTS and live error-free ever after

What is the Devicetree?

What is the Devicetree?

- Tree of devices?



What is the Devicetree?

Source: https://devicetree-specification.readthedocs.io
(trademark and service mark licensed by Linaro Ltd.)

- Data format to describe the non-discoverable hardware
- Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on
  - ACPI-based systems use… ACPI for that!
- DTS - Devicetree Sources are textual representation of Devicetree, to be consumed by the Devicetree compiler (DTC)
- DTB - Devicetree Blob is a compact binary representation of the DTS
- Devicetree bindings - rules how DTS should be constructed and documenting the ABI between software and DTS
- Devicetree schema (DT schema) is the current format of Devicetree bindings
  - … and the toolset
  - … and a project: https://github.com/devicetree-org/dt-schema/

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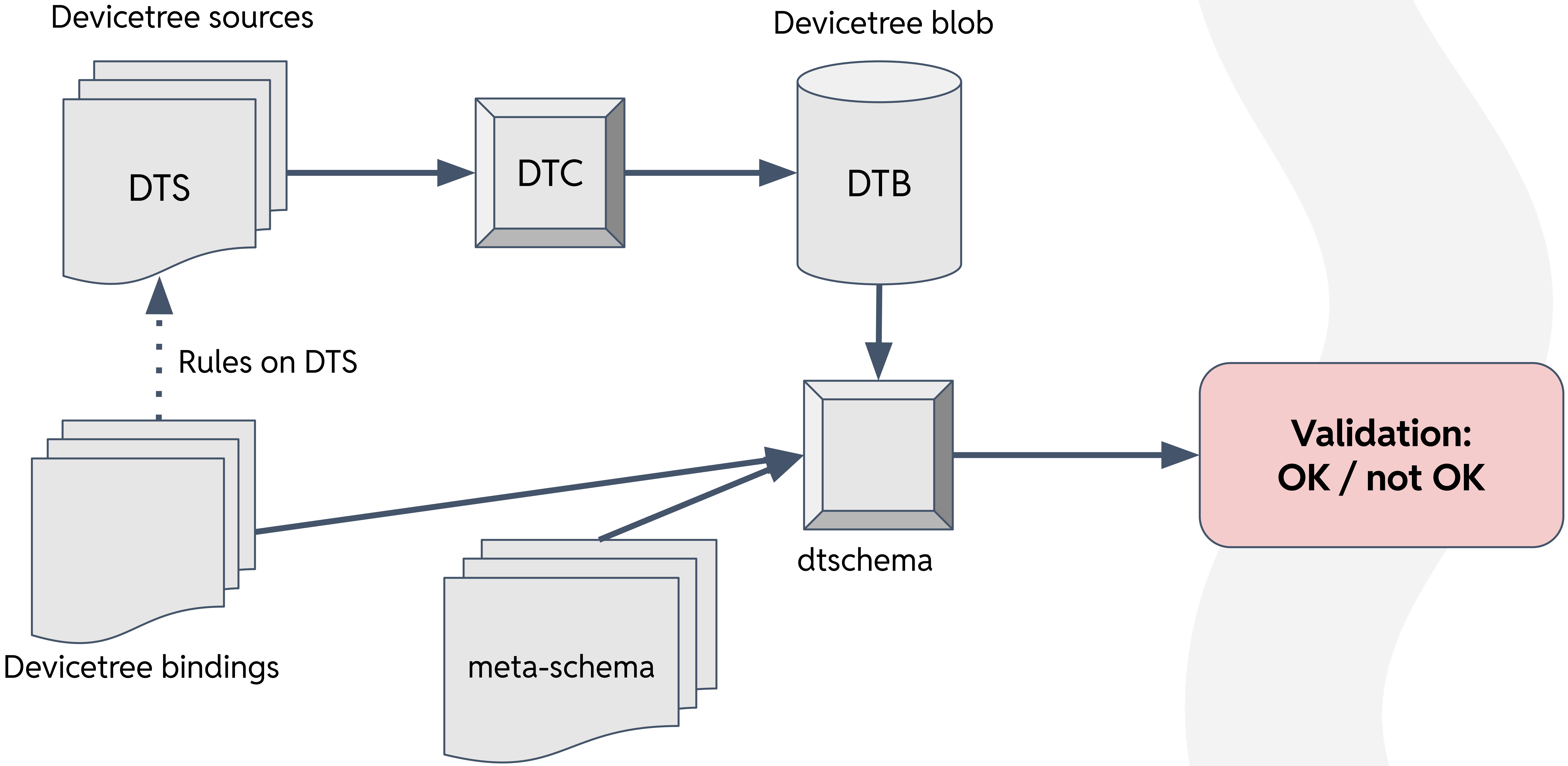

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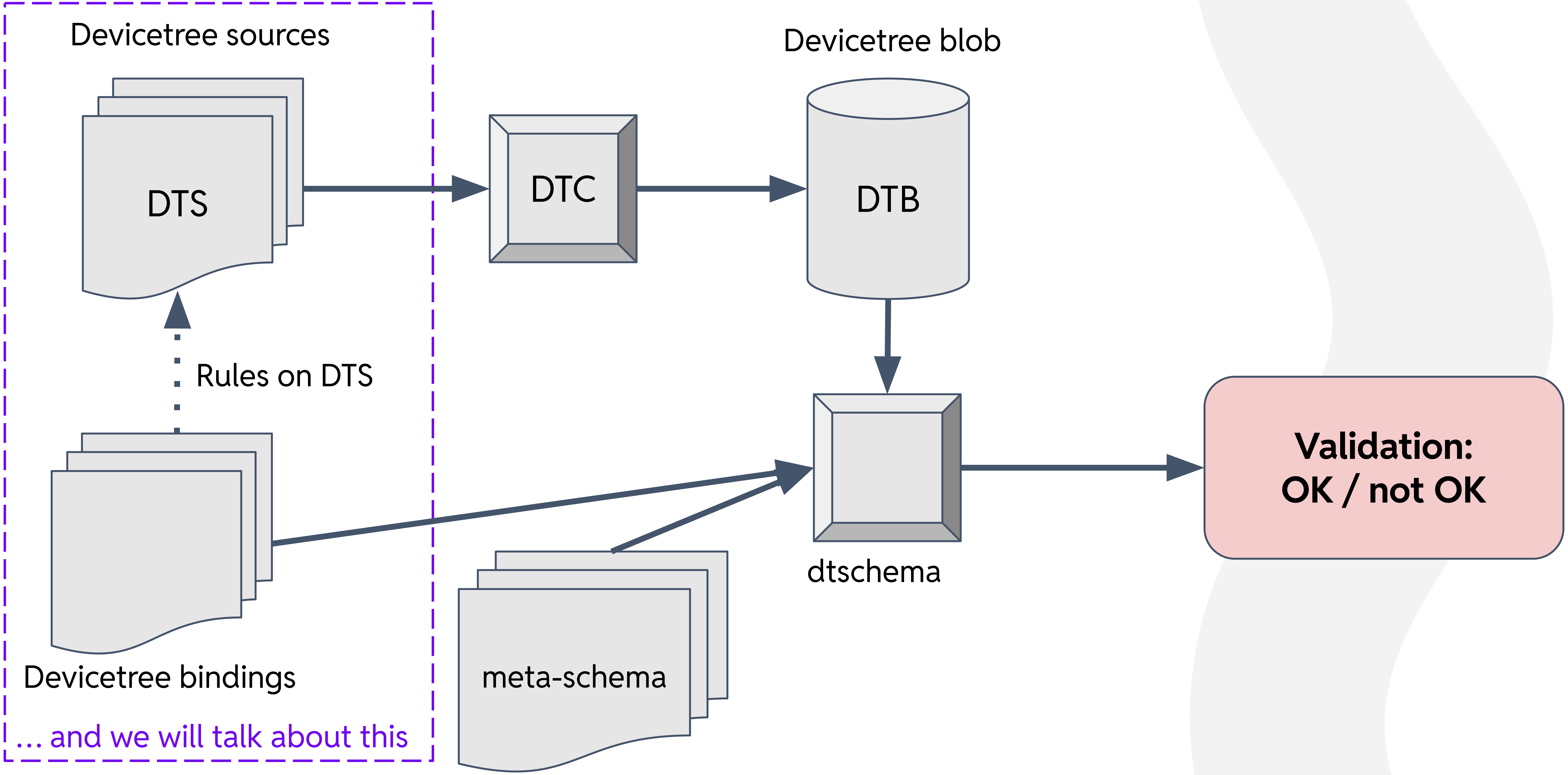

How Does a DTS Look Like?

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How Does a DTS Look Like?

```
/ {
    model = "FooKit";
    compatible = "someone,fookit", "vendor,soc";
    cpus { ... };
    firmware { ... };

    memory@a0000000 {
        device_type = "memory";
        /* We expect the bootloader to fill it */
        reg = <0x0 0xa0000000 0x0 0x0>;
    };

    soc@0 {
        compatible = "simple-bus";
        ranges = <0 0 0 0 0x10 0>;
        ...
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        clock-controller@100000 {
            compatible = "vendor,soc-clock-ctrl";
            reg = <0x0 0x00100000 0x0 0x1f4200>;
            #clock-cells = <1>;
        };

        spi@880000 {
            compatible = "vendor,soc-spi-ctrl";
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- Root node with model and compatible describing the board

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- Root node with model and compatible describing the board
- List of the CPUs
- Some services provided by firmware

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- Total physical memory installed, often placeholder for whatever is provided by the bootloader

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- Root node with model and compatible describing the board
- List of the CPUs
- Some services provided by firmware
- Total physical memory installed, often placeholder for whatever is provided by the bootloader
- I omitted here many more typical device nodes
- Node grouping all MMIO (memory mapped IO) nodes, usually representing the System on Chip
- By convention that’s just a simple-bus
    - simple means: no resources, nothing to do here

How Does a DTS Look Like?

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soc@0 {
    compatible = "simple-bus";
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    spi@880000 {
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```

How Does a DTS Look Like?

- The SoC node has many children representing particular devices
- These devices will have their own compatible and MMIO address
- They might have certain resources (more on that later) and more children

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How Does a DTS Look Like?

- How SPI controller could look like?
- Describes how the child nodes should be addressed
  - One u32 number for address
  - No size for the addresses, why?
  - Because that's SPI Chip Select!

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spi@880000 {
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    touchscreen@0 {
        compatible = "goodix,gt9916";
        reg = <0>;

        interrupts = <162 IRQ_TYPE_LEVEL_LOW>;
        reset-gpios = <&tlmm 161 GPIO_ACTIVE_LOW>;
        avdd-supply = <&vreg_l14b_3p2>;
        spi-max-frequency = <1000000>;

        touchscreen-size-x = <1080>;
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- And some more device-specific properties

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- Device is a Goodix touchscreen
- It has an interrupt line, one reset GPIO
- It cannot operate without power - thus a supply
- And some more device-specific properties
- **Unless you say otherwise with “status” property, each node is enabled by default**

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```

Compatibility

Compatible Property

- Probably the most important property in DTS and bindings: compatible
- List of strings “vendor,device” and usually each string represents one device
  - **No wildcards, no family names**
  - Avoid versions (there are exceptions), but use actual device model name
  - Must be specific: applying to one specific hardware
    - A bit different meaning is for Linux-specific compatibles like syscon or simple-mfd
- It is a list, which means devices could be compatible with each other
  - For example on a Qualcomm SM8750 SoC the GPI DMA controller is:
  - compatible = "qcom,sm8750-gpi-dma", "qcom,sm6350-gpi-dma";

Dedicated, specific front compatible

Fallback compatible

Compatible Devices

- **Each device must have its own, specific compatible**
  - If you use fallback, you need the front compatible
  - If devices are identical, a dedicated front compatible is still necessary
- Compatible tells Linux which driver to bind with the device
- Two devices are compatible when the new device works with Linux drivers bound via fallback (old) compatible
  - Some new features might not be available in such case, that’s fine
  - Dedicated, front compatible might not be ever used by Linux drivers, that’s fine
- If for the new device the **fallback cannot be used by the Linux kernel, devices are most likely not compatible**

How Does DT Binding Look Like

How to Start with a DT Binding?

- Choose the example-schema or some recent, reviewed binding of similar device as a template
- Filename: matching the compatible or one of its fallbacks

```
# SPDX-License-Identifier: (GPL-2.0-only OR BSD-2-Clause)
%YAML 1.2
---
$id: http://devicetree.org/schemas/iio/adc/rohm,bd79104.yaml#
$schema: http://devicetree.org/meta-schemas/core.yaml#

# Short title describing the hardware (not drivers)
title: ROHM Semiconductor BD79104 ADC

# Person(s) caring about this particular hardware (not subsystem maintainers)
maintainers:
  - Matti Vaittinen <mazziesaccount@gmail.com>
```

How to Start with a DT Binding?

```
...
maintainers:
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description:
  Say something more about hardware. Simple devices as such SPI ADC could have
  just one sentence. More complex hardware, some groups of devices or
  anything unusual usually deserves more explanation. Remember about wrapping
  your lines at current coding style (at 80, even though the example here
  for the purpose of presentation does not fit).
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  your lines at current coding style (at 80, even though the example here
  for the purpose of presentation does not fit).

properties:
  # Compatible should be always the first property, see DTS coding style document.
  #
  # Simple cases - one device:
  compatible:
    const: rohm,bd79104
```

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How to Start with a DT Binding?

```
...

properties:
  # Or use enumeration of few incompatible devices, but similar enough to be
  # put in the same binding. Such enumerations or lists are usually ordered
  # alphanumerically:
  compatible:
    enum:
      - rohm,bd79104
      - rohm,bd79115
```

How to Start with a DT Binding?

```
...

properties:
  # Or express compatibility between devices with fallbacks:
  compatible:
    oneOf:
      - const: rohm,bd79104

      # "items" means a list:
      - items:
          # We expect more devices to be compatible with bd79104, thus enumeration:
          - enum:
              - rohm,bd79115
          - const: rohm,bd79104
```

How to Start with a DT Binding?

```
...

properties:
  # Or express compatibility between devices with fallbacks:
  compatible:
    oneOf:
      - const: rohm,bd79104

      # "items" means a list:
      - items:
          # We expect more devices to be compatible with bd79104, thus enumeration:
          - enum:
              - rohm,bd79115
          - const: rohm,bd79104

  # Above maps to DTS, either:
  # 1. compatible = "rohm,bd79104";
  # 2. compatible = "rohm,bd79115", "rohm,bd79104";
```

How to Start with a DT Binding?

```yaml
properties:
  compatible:
    const: rohm,bd79104

  # No need to write descriptions for obvious fields and cases (SPI chip select):
  reg:
    maxItems: 1

  "#io-channel-cells":
    const: 1

  spi-max-frequency:
    maximum: 20000000

  # Regulator supplies, based on datasheet:
  iovdd-supply: true
  vdd-supply: true
```

How to Start with a DT Binding?

```
properties:
  ...

# List which properties are required by the hardware or by drivers:
required:
  - compatible
  - reg
  # Devices rarely operate without power, even if Linux will provide "dummies":
  - iovdd-supply
  - vdd-supply

# This is an SPI device, so needs to reference the schema for SPI devices.
# Often there are other schemas with common properties for devices of similar type.
allOf:
  - $ref: /schemas/spi/spi-peripheral-props.yaml#

# This binding referenced other schema (spi-peripheral-props.yaml) and all properties
# from there apply, thus we want to disallow properties unevaluated so far.
# If we did not reference other schema, this would be "additionalProperties: false".
unevaluatedProperties: false
```

How to Start with a DT Binding?

```
# Examples are used in validation of the binding itself, so it should be complete
# and correct. Do not add more examples than needed.
# Also no need for consumer examples in provider bindings.
examples:
  - |
    spi {
        #address-cells = <1>;
        #size-cells = <0>;

        adc@0 {
            compatible = "rohm,bd79104";
            reg = <0>;
            #io-channel-cells = <1>;
            spi-max-frequency = <4000000>;
            iovdd-supply = <&iovdd_supply>;
            vdd-supply = <&vdd_supply>;
            # Important: No "status" property in examples
        };
    };
```

What Could You Put into DTS?

What Could You Put into DTS?

- TLDR: only hardware
  - or underlying parts of non-discoverable firmware
- Device nodes represent real hardware devices
  - **Not the Linux Device Driver model**
- Properties represent observable hardware characteristics or board wiring
  - No Linux driver choices
    - Usually **don't even reference drivers** in the commit messages or the DT binding description
  - No OS policies
  - No runtime knobs

What Could You Put into DTS?

- I need a child node for my device to instantiate Linux driver (e.g. from MFD driver)
  - NO (usually)
  - Is the hardware piece of your child device re-usable (e.g. RTC block in PMIC)?
  - Does the child node have distinctive resources (e.g. address space, clocks, pins, resets, supplies)?
  - Instead: Fold the child properties into the parent node
- My driver needs to notify others for hot plug events
  - NO
  - Instead: Describe the actual hardware problem
- My driver should not register power supply class for this device
  - NO
  - Instead: Compatible already implies that (e.g. “this variant does not have charger”)
- My driver needs to program register “foo”, so I need “vendor,foo-value”
  - DEPENDS
  - Instead: Please rephrase the problem and the property, to describe the actual hardware feature

What Could You Put into DTS?

- “vendor,allowed-voltages-microvolt”, because my new device, which is compatible with an older one, does not support all voltages
  - NO
  - Instead: Your front, dedicated compatible defines that
- “clock-frequency” property, because I need to configure registers based on some clock
  - NO (usually)
  - Instead: Your device has clock input, thus needs “clocks” property. Your driver should use clk_get_rate().
  - Instead: If you need to configure clock to given frequency: git grep assigned-clock-rates
- I need new “vendor,foo-prop” property
  - WAIT
  - Please look at existing common properties from common schemas or devices representing similar class

What Could You Put into DTS?

- My SoC has two, almost the same PCIe PHYs with slight differences, I need “instance-id” property or custom OF alias to figure out which one is which
    - NO
    - Instead: Maybe other existing, common properties express it, e.g. “num-lanes”
    - Instead: If they have different programming model, then different compatible
    - Instead: Maybe the phandle (cell) arguments from consumer node expresses it, e.g.:

```
phys = <&socphy PHY_TYPE_PCIE>;
```

Validate Your DTS and Live Error-free Ever after

Validate DTS and Bindings

- Validate your binding

```
$ make dt_binding_check DT_SCHEMA_FILES=trivial-devices.yaml
```

- Validate all DTS files in given ARCH against your binding
    - Remember about cross compile options and ARCH=foo
    - This might not show all warnings, because it skips other schema files

```
$ make dtbs_check DT_SCHEMA_FILES=trivial-devices.yaml
```

- Validate one DTS
    - You pass the DTB Makefile target, not DTS

```
$ make CHECK_DTBS=y qcom/sm8450-hdk.dtb
```

- Validate one DTS against your binding
    - You pass the DTB Makefile target, not DTS

```
$ make CHECK_DTBS=y qcom/sm8450-hdk.dtb DT_SCHEMA_FILES=trivial-devices.yaml
```

linaro.org

Thank you

@krzk@social.kernel.org